

DATA ANALYTICS WITH PYTHON

Sales Analysis

Uncovering quarterly sales trends for AAL — a national clothing retailer — using Python, pandas, and matplotlib

Project Overview

Why this analysis was built

AAL is a household clothing brand in the U.S., serving kids, women, men, and seniors from stores across many states. With the business in expansion mode, management wants an in-depth read on sales performance to guide investment decisions.

Data source	Sales.csv — 7,560 rows × 6 columns
Fields	Date, Time, State, Group, Unit, Sales
Tools	Python · pandas · NumPy · matplotlib · scikit-learn
Period	October – December 2020 (7 states, 4 groups)

TASK

“Compile, analyze, and illustrate data findings using charts, graphs, and visual representations to derive meaningful insights and generate comprehensive reports.”

RESULT

A data-driven notebook report that uncovers sales trends at the daily, monthly, state, group, and time-of-day level — to help optimize inventory and marketing.

How the Analysis Was Built

From raw CSV to a full quarterly report

01

Prepare & Clean

Load Sales.csv into a DataFrame ($7,560 \times 6$), confirm zero missing values, and convert Date to a proper datetime.

02

Normalize

Scale Unit and Sales with MinMaxScaler so both range 0.0–1.0 — verified on all 7,560 rows.

03

Visualize Trends

Plot daily Date-vs-Unit and Date-vs-Sales lines for the full quarter, then chunk into Oct / Nov / Dec sub-frames.

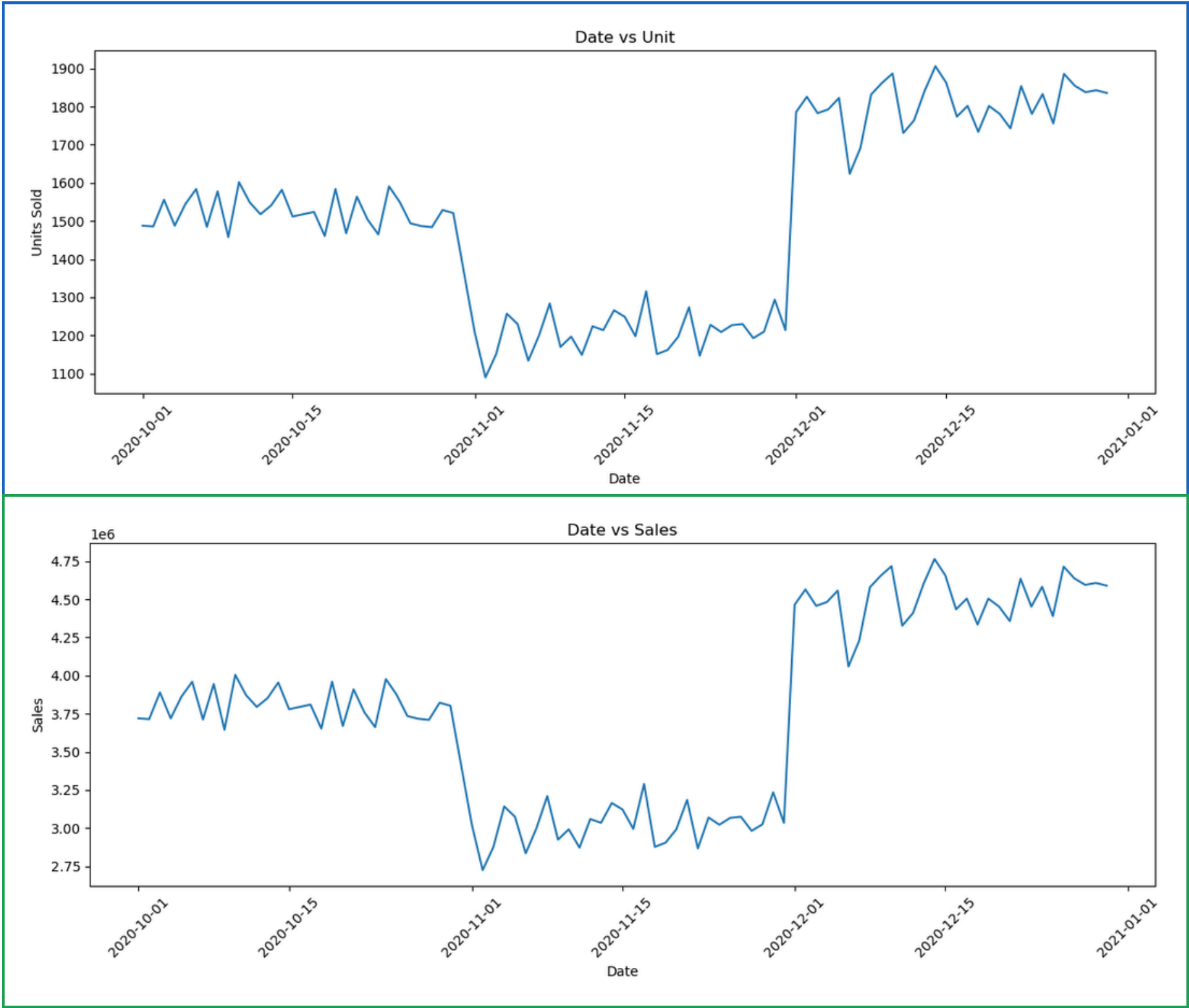
04

Describe & Group

Run describe() overall and per month, boxplot the distributions, then break totals out by state, group, and time of day.

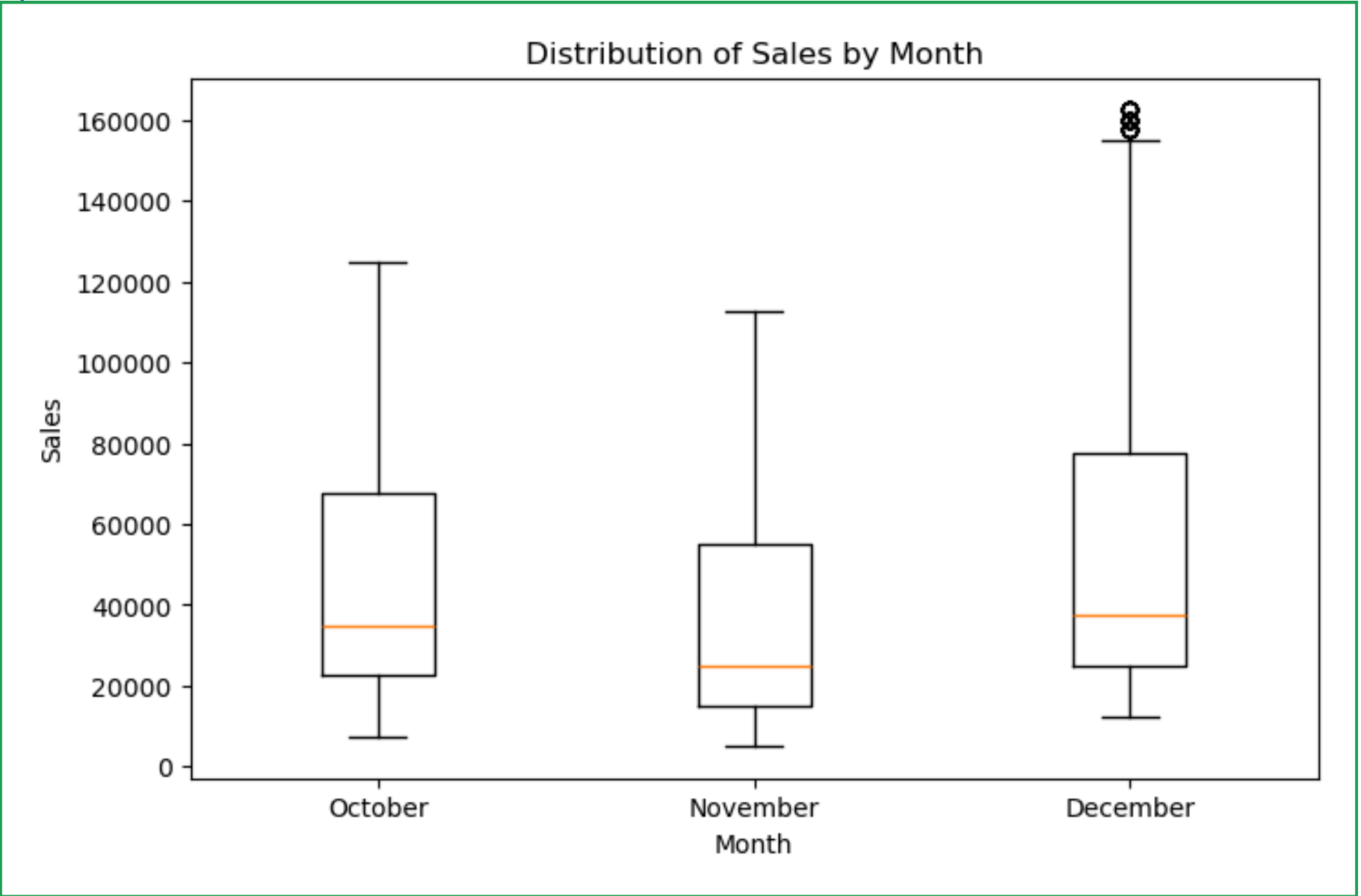
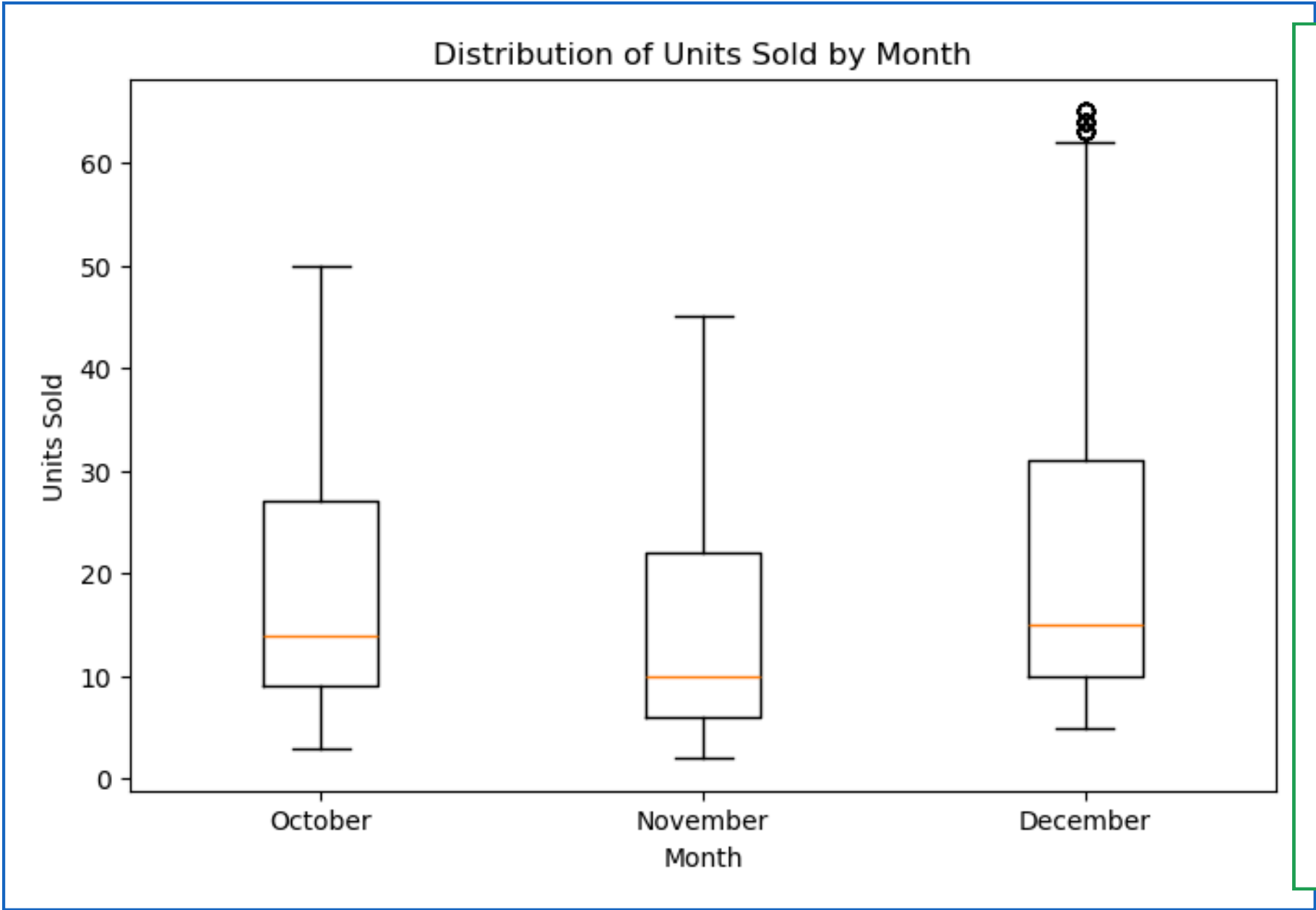
Overall Trend — Units & Sales

Daily totals across the full quarter, straight from the notebook



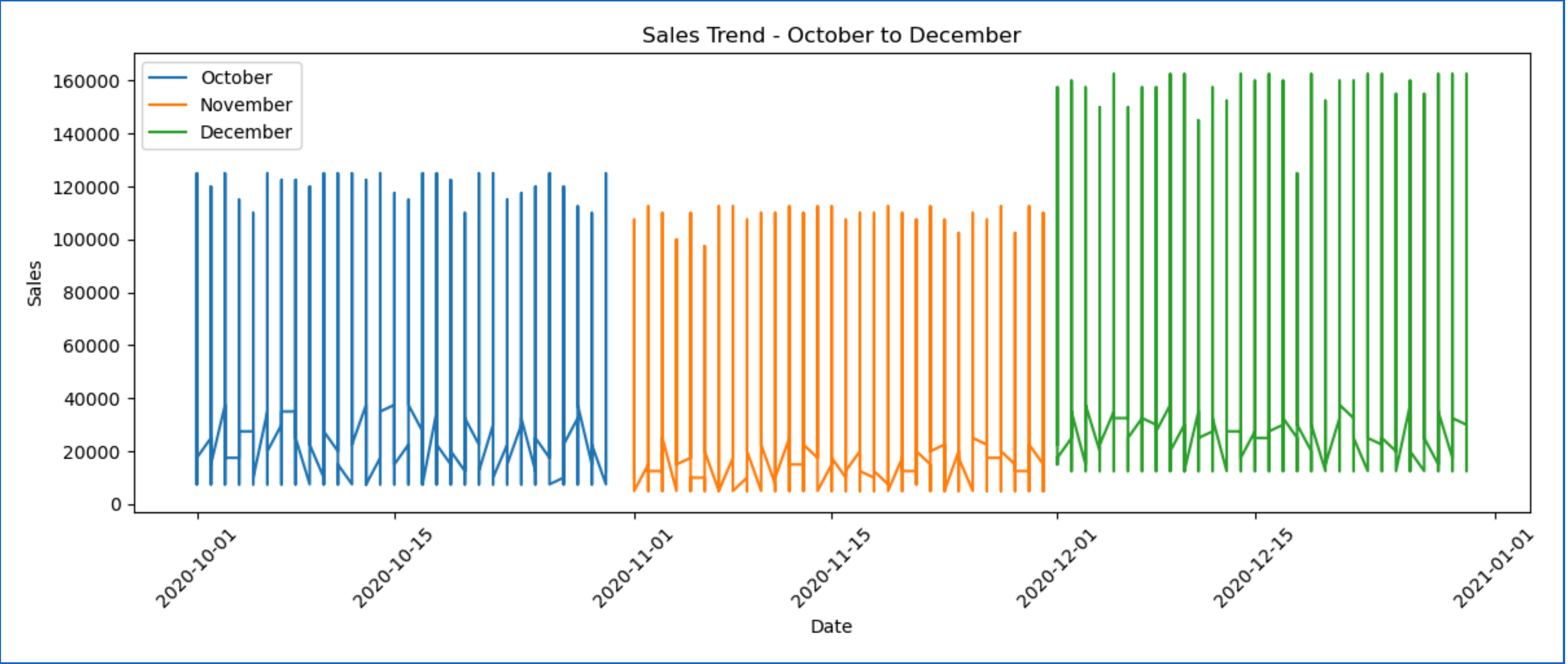
Monthly Distribution

Boxplots of Unit and Sales for October, November, and December



Consolidated Quarterly Trend

October, November, and December overlaid on one sales timeline



Comprehensive Snapshot

Overall and month-by-month totals across the full quarter

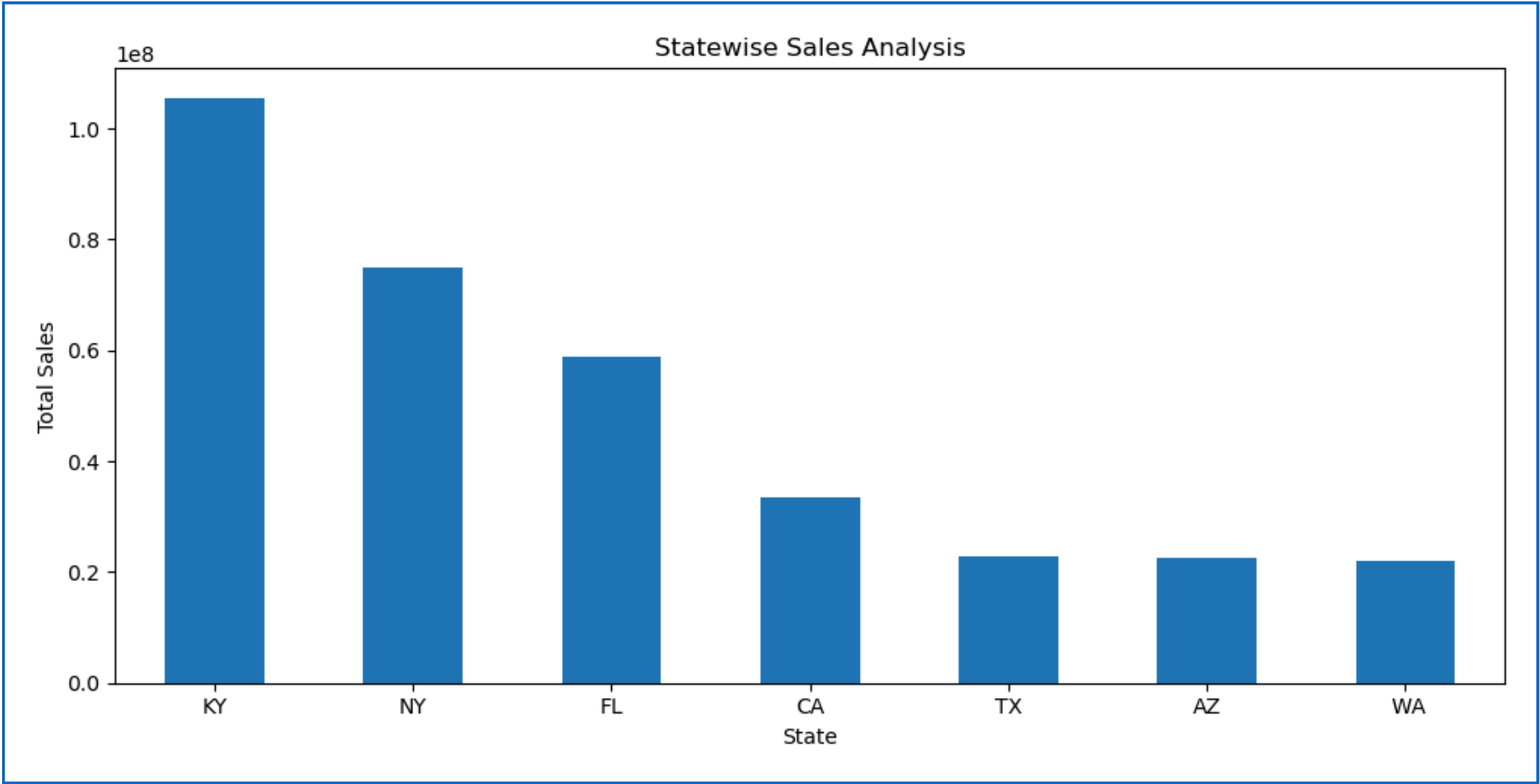
TOTAL UNITS SOLD	TOTAL SALES	AVG. SALE
136,121	\$340,302,500	\$45,013.56

Month	Units Sold	Sales	Share of Quarter Sales
October	45,716	\$114,290,000	33.6%
November	36,273	\$90,682,500	26.7%
December	54,132	\$135,330,000	39.8%

December alone brings in nearly 40% of the quarter's revenue — despite November being the slowest month.

Statewise Sales Analysis

Total sales by state across the full quarter

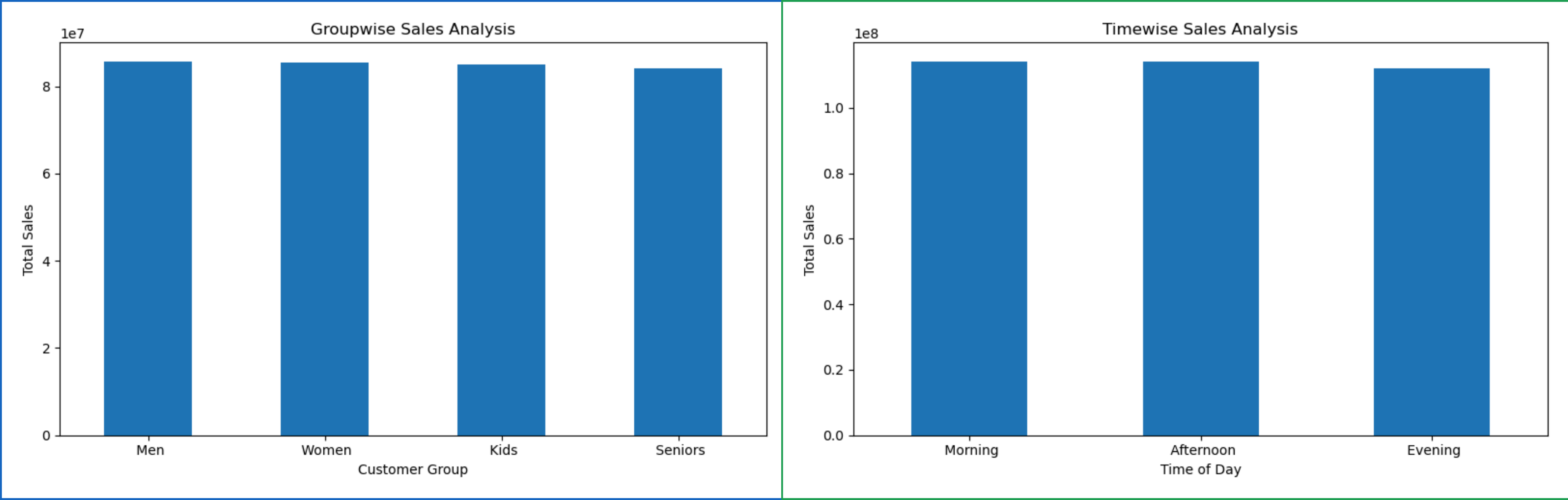


State	Sales
KY	\$105.6M
NY	\$75.0M
FL	\$58.9M
CA	\$33.4M
TX	\$22.8M
AZ	\$22.6M
WA	\$22.2M

Kentucky leads by a wide margin — nearly 5× the smallest market, Washington.

Group & Time-of-Day Analysis

Sales split by customer group and by time of day



Men: \$85.75M · Women: \$85.44M · Kids: \$85.07M · Seniors: \$84.04M | Morning: \$114.2M · Afternoon: \$114.0M · Evening: \$112.1M

Key Insights

What the notebook reveals about AAL's quarter

1 December is the growth engine

Sales jump from an Oct/Nov range of ~\$2.7M–\$4.0M/day to a \$4.3M–\$4.8M/day plateau from Dec 1 onward — the holiday effect.

2 November is the soft spot

Both units (36,273) and sales (\$90.68M) dip to their quarter lows in November, with a sharp trough right after Nov 1.

3 Kentucky is the anchor state

KY alone drives \$105.6M of sales — more than the bottom four states (CA, TX, AZ, WA) combined.

4 Groups and times are remarkably even

Customer groups sit within 2% of each other (\$84M–\$86M), and morning/afternoon/evening are within 2% too — demand isn't segment-driven, it's seasonal.

Recommendations

Turning the analysis into action for AAL

1 Build inventory and staffing plans around the December surge — stock up and staff up well before Dec 1.

2 Investigate November's slump with targeted promotions or bundles to smooth the trough between Q4's two peaks.

3 Protect and reinvest in Kentucky and New York, the two anchor states, while testing growth tactics in the smaller five.

4 Since groups and times are so evenly matched, prioritize seasonal campaigns over group- or time-targeted ones for the biggest lift.